Eureka3.0
Varroc Innovation Hackathon

IGNITE INNOVATION

AUTOMOTIVE INNOVATION HACKATHON

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VARROC eureka CHALLENGE

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Company Website: www.varroc.com

Complete registration and Idea submission at : Eureka Challenge 3.0 - 2026

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Access from your phone/desktop https://unstop.com/competitions/crp-eureka-challenge-30-varroc-1648164

Register Yourself

Log in and subscribe to the challenges.

Give in your creative ideas to the problem statements

Brainstorm and find solution on challenges.

Chance to Meet Varroc Leadership team

Present your idea in big platform

Preplacement offer to shortlist students

Opportunity to work in Core Product development company

Problem Statement 1:

Design a simple, robust and cost-effective retractable footstep for cars and small Commercial Vehicles

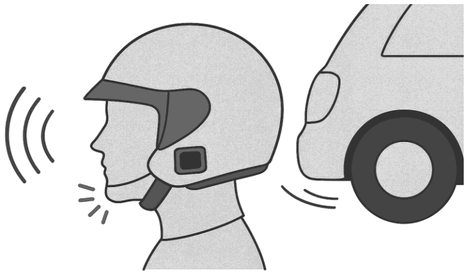
Description : Cars and small commercial vehicles with higher floor heights often use fixed footsteps that reduce ground clearance and are prone to damage on uneven roads. There is a need for a compact and reliable footstep system that provides safe entry and exit without remaining permanently exposed. This project aims to develop a practical retractable footstep mechanism that deploys when door is opened and retracts when its closed, improving accessibility, safety, and vehicle usability.

Scope : 1. Design concept must not use any motor to deploy or retract the footstep.
2. Concepts can be presented in form of CAD models or drawings.
3. Concepts must be supported with detailed analysis.

Note - Concepts would be judged basis novelty, simplicity, form-factor and cost-effectiveness.

Problem Statement 2:

Develop a cost effective and real time horn detection mechanism in a smart helmet



Horn detection mechanism

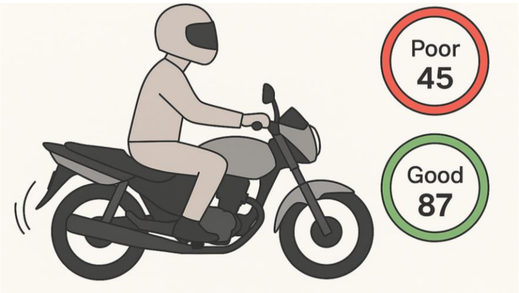
Description : When riders listen to music through helmet-integrated speakers, their ability to hear surrounding vehicle horns is significantly reduced, increasing the risk of accidents. This concept proposes a smart helmet with a horn detection mechanism that identifies external honking sounds and alerts the rider. The system aims to maintain rider awareness and safety without interrupting the listening experience.

Scope : 1. Clear and detailed description of the horn detection mechanism concept must be presented.
2. Concepts must be supported through mathematical calculation, analysis and/or simulations.
3. The mechanism must be conceptualized for the real time operation.

Note - Concepts would be judged basis novelty, simplicity, form-factor and cost-effectiveness.

Problem Statement 3:

Accurate calculation of a driving behavior score on a two-wheeler



Two-wheeler driving behavior score

Description : Two-wheeler riders are highly vulnerable to road risks due to exposure, traffic density, and riding habits. An accurate driving behavior score helps objectively assess riding patterns such as speed control, braking, acceleration, maneuvering, etc. Reliable scoring is essential to identify unsafe behaviors, promote responsible riding, support rider training, and enable fair evaluation for safety programs, insurance benefits, and smart mobility systems.

Scope : 1. Logic behind "how this behavior scoring methodology is accurate and reliable" must be presented.

2. Clear and detailed description of the driving behavior score, sensors used, data logging, etc. must be presented.

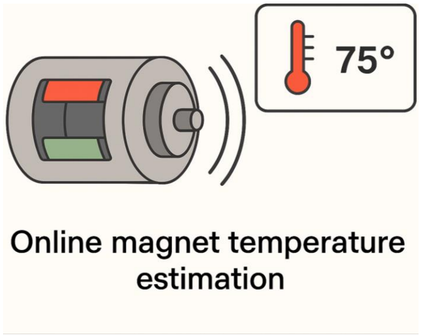
3. Concepts must be supported through mathematical calculation, analysis and/or simulations.

4. The mechanism must be conceptualized for the real time operation.

Note - Concepts would be judged basis novelty, simplicity, form-factor and cost-effectiveness.

Problem Statement 4:

Online direct/indirect measurement/estimation of permanent magnet temperature in PMSM



Description : In Permanent Magnet Synchronous Motors (PMSMs), the temperature of the permanent magnet directly affects motor efficiency, torque capability, and long-term reliability. Excessive magnet temperature can lead to irreversible demagnetization and performance degradation, while direct temperature measurement of the rotor magnet is difficult due to its rotating and enclosed nature. Therefore, accurate online temperature estimation is essential to monitor magnet health in real time, enable safe control strategies, prevent thermal damage, and ensure reliable operation of PMSMs, especially in high-performance and electric vehicle applications

Scope : 1. Clear and detailed description of temperature estimation of permanent magnet, data logging, etc. must be presented.

3. Concepts must be supported through mathematical calculation, analysis and/or simulations.

4. The mechanism must be conceptualized for the real time operation.

Note - Concepts would be judged basis novelty, simplicity, form-factor and cost-effectiveness.

Problem Statement 5:

Moisture removal from closed box

Description : For the given problem, suggest the appropriate type and number of vents or air breathers required for the removal of condensate (condensed moisture) formed due to a sudden drop in outside temperature. The design should ensure that the maximum time required for condensate removal does not exceed 30 minutes. Assume suitable equations/ properties and refer heat transfer books. Refer the vents details mentioned below .

Scope : Excel sheet/ MATLAB code / AI/ML model formulas to quickly decide on condensate formed and breather/vent sections .

- Air Temperature drop from 60 C to 5 C inside the Cubical Box

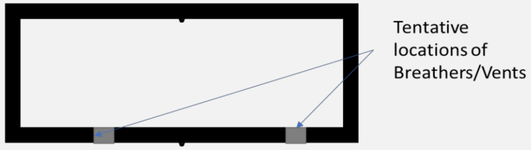
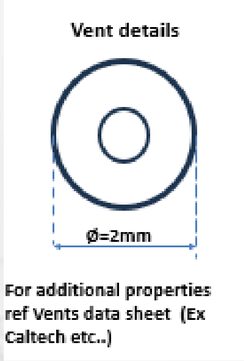


Fig :- Box size = 200 X 100 X 100 mm



Problem Statement 6:

How can automotive lighting systems be optimized for poor road infrastructure and high glare conditions in India

Description : The basic requirement by Automotive lighting systems are “Low beam” for city/village low speed drive & “High Beam” for highspeed driving.

But the in current condition the road-users are using High beams everywhere making other road users like upcoming & ongoing vehicles, pedestrians suffering off glare. This makes uncomfortable situation lands into serious accidents counting on life. Using advanced technologies this concern can be addressed by optimizing the headlamp operating condition

Scope : Effective operating of Main beams of Headlamps for Glare free lighting & to identify or help to identify the poor road conditions earlier; like pat-holes, Unknown obstacles, stray animals, Pedestrians, Road barriers, Diversion barricades.

Problem Statement 7:

How can advanced LED and adaptive lighting technologies be made affordable for mass-market vehicles in India?

Description : LEDs have rapidly replaced halogen and HID technologies across headlamps, DRLs, and rear lamps. Their advantages are well established: higher energy efficiency, longer life, faster response times, compact packaging, and significantly greater design freedom.

Adaptive technologies needs high performance LEDs & the additional precise computing & control system integration on vehicles for operating it. High end software support is another challenge. With such requirements currently its quite costly matter to implement on mass production.

Making it cost down & in the affordable range will allow the OEs to use it on mass production vehicles.

Scope : Efficient design of main beams on Headlamps

Problem Statement 8:

Solution for vehicle cabin overheating in parked condition

Description : With global average temperatures rising and peaks reaching newer records, a parked car, especially for a longer duration under direct sunlight, in tropical regions, essentially becomes as hot chamber. With higher temperatures, come **higher VOCs** (not only from the car but also from the objects inside the car). There have been recorded instances of people forgetting pets & even infants in locked vehicles in some cases. The durability of the under the hood parts is also challenged. There is a need for thermal environment management solutions for parked cars.

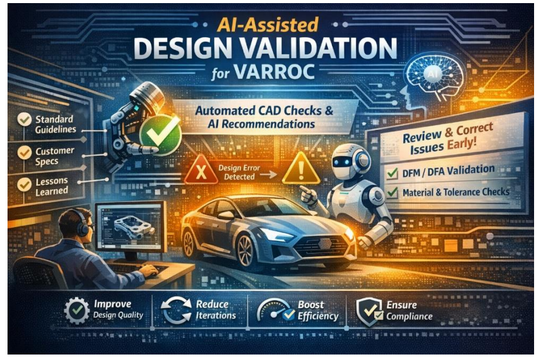
A secondary advantage of this is the reduced energy expense on bringing the cabin temperature down after starting the car.

Scope :

1. Self explanatory working principle for solution on to maintain ambient temperature of car in parked condition.
2. Add schematics / Design concept to explain idea.
3. Explain why your solution is more reliable , cost effective , safe for human & eco friendly sustainable solution .
4. Solution should be provided considering Vehicle is in OFF condition / parked condition (NO battery / NO electric connection available in parked condition)

Problem Statement 9:

AI-Driven Design Intelligence for Early-Stage CAD Validation



Description : In current product development workflows, design engineers rely extensively on manual verification to ensure compliance with established design standards. This process is time-consuming, prone to human error, and highly dependent on individual expertise. As a result, design gaps and non-compliance issues are often identified late in the development cycle, leading to rework, project delays, and reduced overall efficiency.

There is a need for an intelligent system that supports engineers during the design phase by automatically validating designs against standard guidelines and performing real-time checks within the CAD environment. Such an AI-assisted solution should enable early detection of design issues, improve overall design quality, reduce iteration cycles, and ensure accurate product development.

Scope : Design and develop a comprehensive software/Program architecture to achieve the following objectives,
AI-Integrated CAD Assistance:- Development of a module that integrates with CAD tools to analyse designs in real time and provide actionable insights.
Automated Design Validation:- The system should automatically detect deviations, inconsistencies, and potential risks, highlight them within the design, and generate detailed validation reports.

Team Name :

Team members Names	Engineering Specialisation	Graduating year	College Names

Confidential

Tell Us Bit About Yourself

	[Member 1 name]	[Member 2 Name]	[Member 3 Name]
Any projects you've worked on			
Past Hackathon experiences			
Accolades or awards that you have received			
Other details (If any)			

Your Selected Problem Statement

- **Give us an overview of your solution**
- **Process Flow (if applicable)**
- **Uniqueness of your solution**

Design Details

- Market Research and Literature survey
- Flow chart/ block diagrams
- Details with Pictures, Videos of simulations, calculation etc.
- Working Principle
- Simulations/Proto sample
- Benchmarking
- Innovation/improvement over existing solution

Major Risk And Mitigation

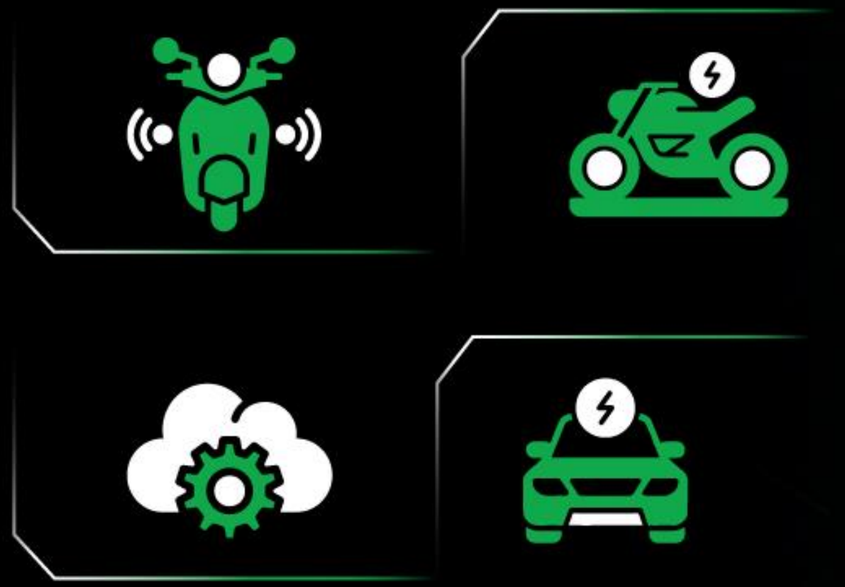
- **Risks to realise the concept**
- **Mitigation**

Business Potential

- **Share details of Business Potential**
- **Market Potential/Demand**

Plan And Budget Needed

- **Major Milestones and Timelines to realize the concept**
- **Budget needed to develop a prototype of this concept**



Thank you

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